Explanation-Guided Active Learning with Controlled TreeSHAP Validation for Static Windows Malware Detection

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Abstract

Static malware detectors require periodic adaptation, but expert labels are costly and probability-margin acquisition ignores how a model assembles its evidence. We evaluate an active-learning family that augments predictive uncertainty with two summaries of signed TreeSHAP contributions: directional entropy of positive versus negative attribution mass and variance across feature coordinates. The evaluation uses an identity-audited subset of EMBER containing 2,000 Windows PE samples and 2,381 features, with five outer folds and five active-learning seeds per fold. At equal budgets, uncertainty plus directional entropy achieved the highest mean ROC and average-precision areas under the learning curve (0.933026 and 0.883336), improving on random acquisition by 0.004249 and 0.014985, respectively, in the same direction in all five folds. The complete three-signal policy required 492 training labels on average to reach a predeclared validation target, 92 fewer than random acquisition, although it did not consistently outperform uncertainty across ranking and low-false-positive criteria. We separately evaluate TreeSHAP as a fitted-model diagnostic across 25 models. Replacing the 50 largest absolute contributions by training means reduced ROC–AUC by 0.306729, versus 0.000979 for equal-budget random replacement; the corresponding median-replacement difference was 0.300119. Rank agreement was moderate and decreased across folds. These results support explanation-guided sample prioritization and relative model reliance under the tested interventions, not causal, executable-level, ransomware-specific, or deployment-level claims.

Index Terms

active learning, explainable artificial intelligence, LightGBM, malware detection, model reliance, portable executable, SHAP, TreeSHAP

1 Introduction

Static machine-learning detectors can score Windows portable executable (PE) files before execution, making them useful for high-volume screening. Public representations such as EMBER summarize byte distributions, strings, headers, sections, imports, exports, and data directories in a fixed vector [1]. Gradient-boosted trees remain strong learners for these heterogeneous attributes and provide efficient, exact attribution mechanisms [2, 3]. Their accuracy nevertheless depends on the labeled distribution used for training. New software, collection sources, packers, and malware families can alter that distribution, while expert analysis of every incoming object is infeasible [4–6].

Pool-based active learning addresses this label bottleneck by selecting a small batch from an unlabeled pool for an oracle. Uncertainty sampling is a natural baseline because it selects samples near the current decision boundary [7, 8]. A probability mar-

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gin, however, collapses the model's evidence into one scalar. Two samples with the same predicted probability may differ substantially: one can have uniformly weak contributions, whereas another can contain large positive and negative effects that nearly cancel. The latter pattern can reveal a region in which the model combines conflicting learned associations, but it is invisible to margin-only selection.

We examine whether signed TreeSHAP summaries provide useful acquisition signals. The proposed family combines predictive uncertainty U with directional entropy H, which measures the balance of positive and negative attribution mass, and signed-attribution variance V, which measures dispersion across feature coordinates. Five policies—Random, Uncertainty, $U + H$, $U + V$, and a complete $U + H + V$ policy—are compared with matched initial sets, model configurations, pools, and label budgets. The evaluation emphasizes the entire learning curve and labels required to reach a predeclared validation target. This avoids using only the final model, for which every policy eventually receives the same full training pool.

Using explanations in acquisition creates a second obligation: the attribution ranking should be tested against fitted-model behavior rather than accepted from a visualization. We therefore conduct a separate controlled TreeSHAP evaluation over 25 fitted models. Sample-specific top-|SHAP| coordinates are replaced by statistics computed only from training data and compared with equal-budget random masks. Mean and median baselines test sensitivity to the replacement rule, while Spearman and Jaccard agreement measure how feature rankings change across model seeds and data folds.

The intervention is intentionally described as a feature-space diagnostic. PE attributes are structured and dependent: replacing isolated histogram, import, header, or section coordinates need not correspond to a valid executable. A large deletion effect therefore supports reliance of the fitted model on the ranked coordinates under a specified intervention. It does not prove that the features cause malicious behavior, that a binary can be edited accordingly, or that the explanation is useful to every analyst. We report heterogeneous diagnostics separately and do not construct an Overall Fidelity Score without a validated normalization and aggregation rule.

The empirical study uses an identity-audited 2,000-sample subset of EMBER with 2,381 static features, 1,500 benign labels, and 500 malware labels. Each row has a source index and SHA-256 identity in a locked manifest. The study uses five outer folds, five active-learning seeds per fold, and five fidelity model seeds per fold. These controls improve on a single-split pilot, but the data remain a small, short-period subset evaluated with random row folds. Consequently, the paper makes no claim about ransomware-specific detection, unseen families, temporal generalization, or production false-alarm rates.

We study three research questions:

- **RQ1:** Under matched training-label budgets, do acquisition policies that combine uncertainty with signed TreeSHAP summaries improve ROC and average-precision learning-curve area over random and uncertainty-only selection?
- **RQ2:** How do the policies trade learning-curve quality, labels required to reach a predeclared validation target, low-FPR behavior, and query computation?
- **RQ3:** Under training-mean and training-median replacement, do top-|SHAP| masks affect fitted-model ranking more than equal-count random masks, and how stable are the rankings across folds and model seeds?

The contributions are:

- An explanation-guided acquisition family that combines prediction margin with directional balance and dispersion of signed TreeSHAP contributions, together with component ablations under matched budgets.
- A multi-fold, multi-seed empirical protocol that reports ROC/AP area under the learning curve (AULC), labels to a validation-defined target, low-FPR summaries, paired fold effects, and computation. The results expose a useful trade-off: $U + H$ has the strongest mean AULC, $U + V$

reaches the target with the fewest labels, and no variant wins every criterion.

- A repeated controlled evaluation of TreeSHAP rankings using two training-derived baselines, equal-budget random controls, three deletion budgets, 25 fitted models, and within-/between-fold stability analysis.
- A claim-bounded reproducibility design that links dataset identity, splits, model seeds, query decisions, predictions, masks, and reported replay checks while separating computational reproducibility from external generalization.

Section 2 reviews static PE learning, batch active learning, and attribution evaluation. Section 3 defines the estimands and claim boundaries. Sections 4 and 5 specify the methods and protocol. Sections 6–8 present results, interpretation, and validity threats; Section 9 concludes.

2 Related Work

2.1 Static PE Malware Learning

Static malware analysis uses evidence available without executing a suspect object. Earlier learned detectors combined hand-engineered program attributes with neural models [9], while raw-byte approaches learned representations directly from complete executables [10]. EMBER established a documented PE feature representation and benchmark [1]; SOREL-20M increased scale and supplied richer metadata [11]. Surveys describe the diversity of representations and learners used in malware analysis [4, 5].

Dataset scale alone does not guarantee realistic evaluation. Random splits can place related families, acquisition sources, time periods, or near-duplicate programs across train and test. Security ML also faces distribution shift and an adaptive adversary, making provenance, split semantics, and operating point central to a credible result [6]. Our study uses a small identity-audited EMBER subset and random outer folds. It improves row-level traceability but does not substitute for temporal, family-disjoint, or external validation.

2.2 Active Learning and Batch Selection

Pool-based active learning alternates model fitting, query selection, oracle annotation, and retraining. Uncertainty sampling is widely used because it is simple and model compatible [7,8]. Its weakness in batch settings is redundancy: the highest-scoring samples can occupy the same region. Core-set acquisition instead seeks representation coverage [12]; ensemble methods query disagreement [13]; BatchBALD uses joint mutual information [14]; and BADGE clusters uncertain gradient embeddings [15].

These baselines are not interchangeable across model classes. BADGE and BatchBALD were designed around differentiable or Bayesian deep models, whereas the present learner is a boosted-tree ensemble. A strong tree-compatible study should nevertheless compare uncertainty plus an explicit diversity constraint and a calibrated committee. The current artifact includes random, uncertainty, and explanation-signal ablations, but not those two stronger families. We treat that omission as an evidence gate before a general superiority claim.

Explanations can support active learning in different ways. Explainable active learning has studied the effect of local explanations on the annotator experience [16]. Our use is algorithmic: TreeSHAP summaries enter the acquisition score before annotation. This does not imply that an attribution with large magnitude or balanced signs yields high expected information gain. It defines a falsifiable heuristic whose value must be established against matched baselines.

2.3 Attribution and Fidelity Evaluation

LIME approximates a local decision surface [17]; SHAP unifies additive attributions through Shapley values [18]; and TreeSHAP specializes the computation to tree ensembles [3]. Explanation evaluation is multi-dimensional. Sanity checks can expose explanations that do not respond to model parameters [19], and adversarial constructions can manipulate post-hoc explanations [20]. Security data add constrained and attacker-controlled inputs [21].

Deletion tests measure whether intervening on highly ranked coordinates changes model output. Their interpretation depends on the replacement distribution: feature removal can create out-of-distribution inputs and confound the ranking comparison [22, 23]. Training-derived mean and median baselines avoid treating zero as universally absent, but they do not preserve PE structure. Equal-count random masks control the nominal intervention budget but may frequently choose coordinates unused by a tree model. Stronger controls include used-feature random selection, magnitude matching, group-aware replacement, and attribution permutation.

The contribution is therefore not the broad claim that combining active learning and explanations is unprecedented. It is the specific signed-mass acquisition construction and its repeated, claim-bounded evaluation together with a controlled assessment of the attribution ranking used by the policy.

3 Problem Formulation and Claim Boundaries

3.1 Pool-Based Learning

Let $x \in \mathbb{R}^d$ be a static PE feature vector and $y \in \{0, 1\}$ its benign/malware label. At acquisition round r, the labeled and unlabeled portions of the development pool are $\mathcal{L}_r$ and $\mathcal{U}_r$. Model f_r is fitted only on $\mathcal{L}_r$ and assigns probability-like score $p_i = f_r(x_i)$. Acquisition function $Q_r(x)$ ranks $x \in \mathcal{U}_r$; the oracle reveals labels for the highest-ranked batch, which is moved to $\mathcal{L}_{r+1}$.

Within each outer fold and seed, all policies share the same initial labeled set, remaining pool, validation data, test data, model hyperparameters, and cumulative budget. Test outcomes do not influence queries, target definition, threshold selection, or stopping. The oracle is simulated by revealing retained labels. A training-label budget is thus a count of oracle queries, not measured analyst time or total dataset annotation cost.

3.2 Learning-Curve Estimands

Every run produces ROC–AUC and average precision (AP) at budgets $B \in \{200, 300, \ldots, 1200\}$. The normalized area under a metric learning curve $m(B)$ is

$$\mathrm{AULC}(m) = \frac{1}{B_{\max} - B_{\min}} \int_{B_{\min}}^{B_{\max}} m(B)\, dB, \quad (1)$$

computed by trapezoidal integration over measured budgets. AULC summarizes how quickly useful ranking is acquired; the endpoint is not discriminative when all policies eventually train on the same full pool.

For each outer fold, a full-pool reference model defines a validation target

$$t_{\mathrm{fold}} = 0.5 + 0.95 \left(\mathrm{AUC}_{\mathrm{val}}^{\mathrm{full}} - 0.5 \right). \quad (2)$$

Labels-to-target is the first training budget whose validation AUC reaches this level. The full-pool reference and validation labels are additional experimental resources and must be reported separately from queried training labels.

Low-FPR summaries include normalized partial AUC up to FPR 5% and TPR learning curve area at FPR caps of 1% and 5%. These quantities describe the measured outer folds. They do not estimate enterprise false-alarm rates near 10^{-3}–10^{-4}, for which substantially more benign test samples are required.

3.3 Explanation Object

For a fitted tree model and sample x_i, let $\boldsymbol{\phi}_i = (\phi_{i1}, \ldots, \phi_{id})$ be the signed TreeSHAP contribution vector and ϕ_{i0} the bias. Native TreeSHAP is checked on the raw-score scale:

$$f_{\mathrm{raw}}(x_i) = \phi_{i0} + \sum_{j=1}^{d} \phi_{ij}. \quad (3)$$

Numerical reconstruction verifies the implementation but not semantic feature identity, causality, or analyst usefulness.

The deletion diagnostic selects sample-specific mask $M_i^{(k)}$ and replaces selected coordinates by a

Table 1: Positioning relative to representative acquisition and attribution-evaluation methods. "Required extension" denotes a comparison that is not present in the retained confirmatory report.

Study or family	Principal signal	Explicit batch diversity	Explanation role	Relationship to this work
Cohn *et al.* [7] / uncertainty	Model uncertainty or error reduction	Method dependent	None	Measured baseline
Core-set [12]	Representation coverage	Yes	None	Required extension
Ensemble disagreement [13]	Predictive disagreement	Indirect	None	Required extension
BatchBALD [14]	Joint mutual information	Yes	None	Requires tree-compatible adaptation
BADGE [15]	Uncertain gradient embeddings	Yes	None	Requires model-compatible adaptation
XAL [16]	Queried samples plus local explanations	Not central	Annotator-facing	Different explanation role
Security XAI evaluation [21]	Explanation behavior under security constraints	Not applicable	Evaluation	Motivates controlled diagnostics
This work	Margin plus TreeSHAP signed-mass summaries	No	Acquisition and post-hoc diagnostic	Measured vs. random/uncertainty; diversity remains open

training-derived baseline b_j:

$$\widetilde{x}_{ij} = \begin{cases} b_j, & j \in M_i^{(k)}, \\ x_{ij}, & \text{otherwise.} \end{cases} \tag{4}$$

For top-SHAP deletion, $M_i^{(k)}$ contains the k largest $|\phi_{ij}|$. The control contains k uniformly selected coordinates. Baseline b_j is either the training mean or median. This intervention measures fitted model reliance in feature space; it need not correspond to a valid PE binary.

3.4 Security Scope

The measured label is generic Windows PE malware. Original SHA-256 identities support traceability, but the study does not establish ransomware labels, family-held-out performance, or zero-day detection. The adversarial scope also excludes a white-box attacker who optimizes pool membership, labels, or features against the acquisition rule. These limits are part of the estimand rather than footnotes to a broader deployment claim.

4 Methodology

4.1 Static Learner and Study Workflow

LightGBM represents the raw score as an additive ensemble

$$F(x) = \sum_{m=1}^{M} \eta h_m(x), \tag{5}$$

where h_m is a decision tree, M is the number of boosting rounds, and η is the learning rate [2]. Its histogram-based learner handles heterogeneous PE attributes efficiently. Native TreeSHAP supplies signed per-coordinate contributions without fitting a separate local surrogate [3].

Figure 1 separates three data flows. The active-learning loop trains on the current labeled set, scores only the remaining development pool, queries a batch, and retrains. Validation defines the target and any operating threshold. The fixed outer test set is used only for reporting. A separate fidelity path trains controlled models on the outer-fold development data and intervenes on test vectors. Fidelity results are not treated as a causal explanation for active-learning gains.

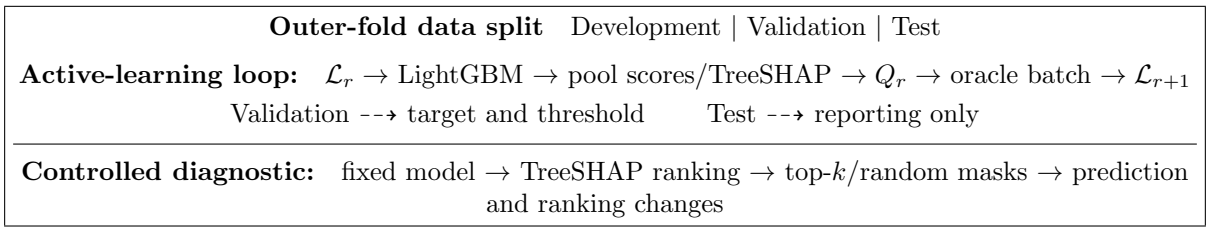


Figure 1: Study workflow. Solid arrows form the acquisition loop. Dashed arrows denote evaluation inputs that never enter query selection. The controlled TreeSHAP diagnostic is fitted and interpreted separately.

4.2 Predictive Uncertainty

For predicted probability p_i, uncertainty is

$$U_i = 1 - 2|p_i - 0.5|. \tag{6}$$

It equals one at the decision boundary and zero at either probability extreme. This score is computationally inexpensive and supplies the strongest measured baseline beyond random selection.

4.3 Directional Entropy

For signed attribution vector $\boldsymbol{\phi}_i$, define positive and negative masses

$$A_i^+ = \sum_j \max(\phi_{ij}, 0), \qquad A_i^- = -\sum_j \min(\phi_{ij}, 0). \tag{7}$$

If $A_i^+ + A_i^- > 0$, the positive-mass share is

$$q_i = \frac{A_i^+}{A_i^+ + A_i^-}. \tag{8}$$

Directional entropy is

$$H_i = -q_i \log_2 q_i - (1 - q_i) \log_2 (1 - q_i), \tag{9}$$

with $H_i = 0$ for an all-zero or one-sided vector. H_i is high when substantial positive and negative attribution masses coexist. It is not entropy of the predicted class distribution and not calibrated epistemic uncertainty.

Let $S_i = \sum_j \phi_{ij}$ and $L_i = \|\boldsymbol{\phi}_i\|_1$. Because $A_i^+ - A_i^- = S_i$ and $A_i^+ + A_i^- = L_i$,

$$q_i = \frac{1}{2} + \frac{S_i}{2L_i}, \qquad L_i > 0. \tag{10}$$

Thus, H_i is a signed-mass balance statistic determined by the raw-score contribution and L_1 attribution magnitude. The identity motivates direct attribution-norm baselines in future confirmatory extensions.

4.4 Signed-Attribution Variance

The dispersion signal is

$$V_i = \frac{1}{d} \sum_{j=1}^{d} (\phi_{ij} - \bar{\phi}_i)^2, \qquad \bar{\phi}_i = \frac{1}{d} \sum_j \phi_{ij}. \tag{11}$$

Expanding the centered square gives

$$V_i = \frac{\|\boldsymbol{\phi}_i\|_2^2}{d} - \frac{S_i^2}{d^2}. \tag{12}$$

It therefore summarizes coordinate dispersion of one explanation; it is not variance across trees, posterior samples, or independently fitted models.

4.5 Composite Acquisition Policies

Because U, H, V have different numeric ranges, each signal S is min–max normalized over the current pool:

$$\widetilde{S}_i = \begin{cases} \dfrac{S_i - S_{\min}}{S_{\max} - S_{\min}}, & S_{\max} > S_{\min}, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

The general score is

$$Q_i = \alpha \widetilde{U}_i + \beta \widetilde{H}_i + \gamma \widetilde{V}_i. \tag{14}$$

Table 2: Measured acquisition policies.

Policy	(α, β, γ)
Random	not applicable
Uncertainty	$(1, 0, 0)$
$U + H$	$(3/7, 4/7, 0)$
$U + V$	$(0.5, 0, 0.5)$
XAI-Full	$(0.3, 0.4, 0.3)$

Table 2 defines the measured policies. Random assigns an independent random score; ties in deterministic policies are resolved using the retained row identifier.

Min–max scaling is sensitive to pool extremes, and the weights were not selected through an independent outer optimization study. The top-100 rule also lacks an explicit diversity constraint. These are tested configurations, not established optima. Sensitivity, uncertainty-plus-diversity, and calibrated committee comparisons remain required before claiming dominance over strong acquisition families.

4.6 Controlled TreeSHAP Diagnostics

For $k \in \{10, 25, 50\}$, each test sample receives a top-|SHAP| mask and 20 uniform random masks of equal size. The intervention in (4) is repeated with the training-column mean and median. Primary outputs are the post-mask ROC–AUC and decrease relative to the unmasked model. The comparison of interest is

$$\Delta_k^{\mathrm{rel}} = \Delta\mathrm{AUC}_k^{\mathrm{top}} - \Delta\mathrm{AUC}_k^{\mathrm{random}}. \tag{15}$$

Absolute differences are preferred to ratios because random-mask decreases are near zero.

Global ranking stability is measured using Spearman correlation. Set agreement for the highest-ranked k coordinates uses

$$J_k(A, B) = \frac{|T_k(A) \cap T_k(B)|}{|T_k(A) \cup T_k(B)|}, \tag{16}$$

where $T_k(A)$ and $T_k(B)$ are top-k feature sets of two fitted models. Comparisons are separated into different seeds within the same fold and models from different folds.

Explanation consistency (EC) and contrastive geometry (CF) are retained only as descriptive diagnostics. EC correlates pairwise cosine geometry in standardized feature and signed-attribution spaces. CF normalizes the distance between label-conditioned SHAP centroids by the largest sample attribution norm. A common permutation of attribution columns preserves additivity, pairwise cosines, centroid distance, and norms while assigning effects to incorrect feature names. Consequently, additivity, EC, and CF cannot certify semantic correctness, and no aggregate OFS is constructed.

5 Experimental Protocol

5.1 Data Identity and Provenance

The retained study report identifies an EMBER-derived table with 2,000 rows, 2,381 static features, 1,500 benign labels, and 500 malware labels. All `row_id`, source indices, and sample SHA-256 values are unique, and the audit found no exact duplicate feature vectors. Samples appeared during November–December 2018; AVClass metadata exists for 498 malware rows but is not treated as ransomware ground truth.

The locked dataset SHA-256 is `69f0ae680846edd95458d1e7aaabe36f8-866f4f22a6d11488b126ed026b28963`; the manifest SHA-256 is `a2ca16402653e6cc51fdd7a4857d007faf86807b044d78654f180409c73565bb`. The subset was recovered from available EMBER test material rather than sampled according to the complete official train/test protocol. We therefore call it an identity-audited internal subset, not a new official EMBER benchmark.

5.2 Outer-Fold Separation

Five outer folds define disjoint test partitions. Within each outer development partition, retained labels are separated into the active-learning pool and validation resources. The report records that test rows never enter query selection, target construction, threshold selection, or model selection. All policies within a

Table 3: Dataset and evaluation units.

Item	Value
Samples	2,000
Static features	2,381
Benign / malware	1,500 / 500
Outer folds	5
AL seeds per fold	5
Fidelity model seeds per fold	5
Initial AL labels	200
Query rounds / batch	10 / 100
Maximum training labels	1,200
Fidelity deletion budgets	10, 25, 50
Random masks per model/budget	20

fold and AL seed receive the same initial labeled rows and remaining pool. The unit for paired policy comparison is the outer fold after averaging its five seed trajectories; 25 fold–seed runs are not treated as 25 independent test datasets.

Random row folds do not prevent related families, acquisition sources, or time periods from crossing partitions. SHA uniqueness and absence of exact feature duplicates address only exact identity leakage.

5.3 Active-Learning Configuration

The AL seeds are 11, 22, 33, 44, and 55. Each run begins with 200 stratified training labels and performs ten rounds of 100 queries, giving budgets from 200 through 1,200. LightGBM uses 60 trees, learning rate 0.05, 15 leaves, maximum depth 4, and deterministic seed-specific training. At each round, the runner saves model configuration, pool membership, query components, selected row IDs, validation/test scores, and metrics.

The full-pool reference in each fold sets the target in (2). This makes target attainment comparable within that fold but introduces additional validation and reference-model cost. The study reports training labels to target and does not translate them into analyst hours.

Query timing covers pool prediction, TreeSHAP computation where applicable, additivity checking, score normalization, and query construction as recorded by the runner. It does not include human annotation, data acquisition, endpoint feature extraction, or production inference.

5.4 Fidelity and Stability Configuration

For each outer fold, five LightGBM models use seeds 41–45, 100 trees, learning rate 0.1, feature fraction 0.8, bagging fraction 0.8, 15 leaves, and maximum depth 4. Controlled subsampling creates variation across fitted models. Each of the 25 models is evaluated under both mean and median replacement. Three top-k and 3×20 random-mask batches are generated for each replacement baseline, yielding 50 model–baseline combinations.

The retained report records replay of 3,150 mask batches and 1,260,000 prediction records, with 50/50 model–baseline verification cases passing. Active-learning artifacts record 1,375 checkpoints, 1,100,000 prediction records, and 125,000 query records, with all five folds passing the reported verification. These are reported properties of the retained artifact. The experiment commits are not present in the cloud checkout used to assemble this manuscript revision, so a fresh hash check and replay remain mandatory before submission.

5.5 Metrics and Statistical Reporting

ROC–AUC and AP measure ranking [24,25]. AULC, labels-to-target, normalized low-FPR summaries, and computation describe the AL trajectory. Paired effects are calculated within outer fold. Direction counts such as 5/5 or 4/5 folds are reported without treating five folds as sufficient for a strong asymptotic significance claim.

Fidelity results aggregate across fitted models and distinguish seed variation within a fold from variation across folds. Random masks estimate control-mask variation conditional on the retained fitted models. Neither the mask repetitions nor pairwise stability comparisons are independent draws from a deployment population.

6 Results

6.1 RQ1: Learning-Curve Quality

Table 4 reports performance integrated across the full training budget. $U + H$ obtained the highest mean ROC-AULC (0.933026) and AP-AULC (0.883336). Relative to Random, its paired fold differences were +0.004249 in ROC-AULC and +0.014985 in AP-AULC, with the same favorable direction in all five outer folds. The gains over Uncertainty were smaller: +0.001754 ROC-AULC and +0.001401 AP-AULC, favorable in four of five folds.

$U + V$ did not lead AULC, but it reached the validation target with the smallest mean training budget, 488 labels. XAI-Full reached the target at 492 labels. The latter required 92 fewer labels than Random, a 15.8% reduction under this target definition, and 28 fewer than Uncertainty. $U + H$ required 76 fewer labels than Random (13.0%). These are oracle-query counts under retained labels; validation labels and full-pool reference training are additional resources.

The three explanation-guided variants therefore express different trade-offs. $U + H$ produces the strongest average learning trajectory, $U + V$ reaches the declared target earliest, and XAI-Full balances the signals without leading every criterion. Figure 2 visualizes this separation. The finding answers RQ1 positively relative to random acquisition, but only conditionally relative to uncertainty sampling.

Table 5 makes the strongest limitation explicit. XAI-Full improved ROC-AULC only in three of five fold comparisons against Uncertainty, and its mean AP-AULC difference was slightly negative. The three-signal combination is thus not an empirically universal winner. The clearest evidence for explanation-derived acquisition comes from $U + H$ relative to Random and, more modestly, relative to Uncertainty.

6.2 RQ2: Low-FPR Behavior and Computation

Table 6 reports normalized partial-AUC and TPR learning-curve summaries. All explanation-guided methods improve substantially over Random. Against Uncertainty, however, differences are small and change sign. Uncertainty has the highest pAUC at FPR$\leq$ 5% and TPR-AULC at the 1% cap; XAI-Full is highest at the 5% cap. $U+H$ is slightly below Uncertainty at the 1% cap and slightly above it at 5%.

The operating-region result changes the practical interpretation. Explanation signals improve sample efficiency relative to random annotation, but the retained study does not establish lower alert burden than a well-implemented uncertainty baseline. Each outer test fold contains only about 300 benign samples. One false positive therefore changes FPR by roughly 0.33 percentage points, making much lower operational regions unresolvable.

Explanation scoring increases computation. $U+H$ adds approximately 0.227 s of query processing across ten rounds relative to Uncertainty and increases total model-plus-query time by 7.93%. XAI-Full increases the total by 4.86%; query processing constitutes about 5.42% of its reported model-plus-query time. These small absolute values apply to a 2,000-sample experiment and exclude annotation, data loading outside the timed scope, and endpoint detection latency.

RQ2 therefore has no single optimal policy. $U+H$ is preferred when integrated ranking quality is primary, $U+V$ when target attainment under the measured rule is primary, and Uncertainty remains competitive in the low-FPR summaries while avoiding TreeSHAP overhead.

6.3 RQ3: Controlled Deletion

Table 7 reports AUC decreases averaged over the retained fidelity models. Under both replacement baselines, top-|SHAP| deletion has a large, monotonic effect as k increases, whereas uniform random deletion has a near-zero mean effect. For mean replacement, the top-minus-random difference is 0.047617 at $k = 10$, 0.135323 at $k = 25$, and 0.305750 at $k = 50$. Median replacement yields comparable differences of 0.053388, 0.124854, and 0.300119.

Figure 3 shows that the result is not specific to mean replacement. It establishes relative fitted-

Table 4: Active-learning performance, target attainment, and computation. AULC integrates the trajectory over 200–1,200 training labels. Labels-to-target excludes validation labels and the full-pool reference cost.

Policy	ROC-AULC	AP-AULC	Labels to target	Query, 10 rounds (s)	Model + query (s)
Random	0.928777	0.868350	584	0.0194	4.1947
Uncertainty	0.931272	0.881934	520	0.0191	4.1454
$U + H$	**0.933026**	**0.883336**	508	0.2459	4.4741
$U + V$	0.930377	0.880164	**488**	0.2298	4.3826
XAI-Full	0.932202	0.881638	492	0.2358	4.3471

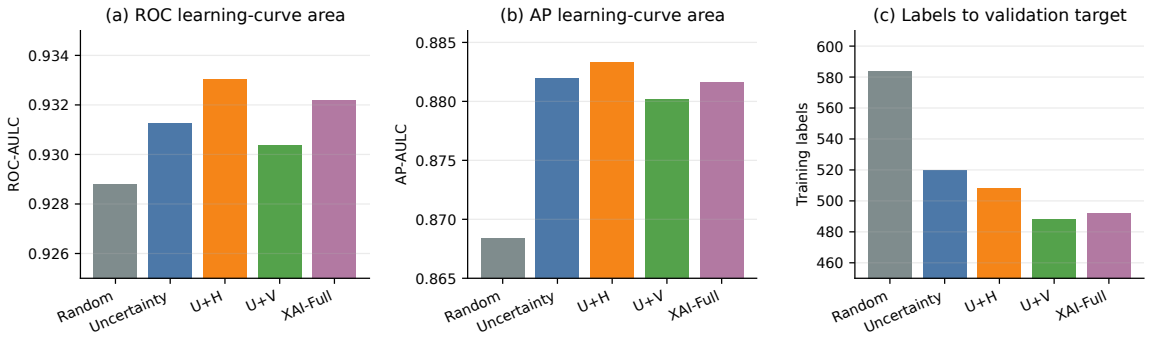


Figure 2: Confirmatory active-learning summaries. Higher AULC is better; fewer training labels to the predeclared validation target are better. Error bars are omitted because the readiness snapshot supplies aggregate means and paired fold effects rather than the underlying fold-level vectors in this checkout.

Table 5: Selected paired outer-fold effects.

Comparison	ΔROC	ΔAP	Folds
$U + H$ vs. Random	+0.004249	+0.014985	5/5
$U + H$ vs. Uncertainty	+0.001754	+0.001401	4/5
XAI-Fullvs. Random	+0.003425	+0.013287	5/5
XAI-Fullvs. Uncertainty	+0.000931	−0.000297	3/5

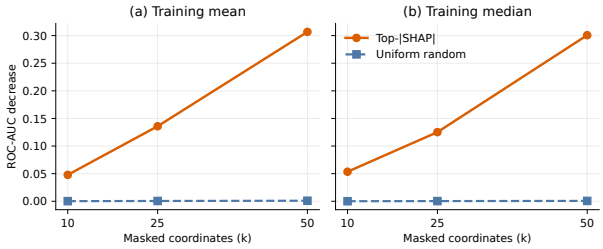


Figure 3: ROC–AUC decrease under top-|SHAP| and uniform random replacement. Absolute effects are emphasized because ratios are unstable when the random-control decrease is near zero.

model reliance under both tested training-derived baselines. Equal coordinate count does not match perturbation magnitude or ensure that random coordinates participate in the trees; the result should therefore not be called an executable-level fidelity certificate.

6.4 RQ3: Attribution-Rank Stability

Table 8 separates model-seed and fold changes. Within a fixed fold, global Spearman agreement is 0.6144; top-10 Jaccard agreement is 0.6221 and decreases to 0.4901 at top 50. Across folds, Spearman agreement falls to 0.5512, and top-k Jaccard values range from 0.4416 to about 0.30.

Table 6: Learning-curve summaries in restricted false-positive regions. These are not deployment estimates at FPR 10^{-3} or below.

Policy	pAUC, FPR$\leq$ 5%	TPR-AULC, FPR cap 1%	TPR-AULC, FPR cap 5%
Random	0.804096	0.580620	0.731980
Uncertainty	**0.826300**	**0.638180**	0.769260
$U + H$	0.824774	0.636780	0.770620
$U + V$	0.825194	0.626260	0.768140
XAI-Full	0.823525	0.631860	**0.771460**

Table 7: Controlled feature-space deletion across 25 fitted models. Values are ROC–AUC decreases from the corresponding unmasked model.

Replacement	k	Top-SHAP drop	Random drop	Top minus random
Mean	10	0.047817	0.000200	0.047617
Mean	25	0.135853	0.000530	0.135323
Mean	50	0.306729	0.000979	0.305750
Median	10	0.053483	0.000094	0.053388
Median	25	0.125180	0.000326	0.124854
Median	50	0.300777	0.000658	0.300119

Table 8: TreeSHAP ranking stability.

Comparison	ρ	J_{10}	J_{25}	J_{50}
Seed, same fold	0.6144	0.6221	0.5398	0.4901
Different fold	0.5512	0.4416	0.3073	0.3020

The rankings are neither arbitrary nor invariant. They exhibit moderate agreement, with data partitioning exerting more influence than controlled model randomness. This finding is important for use in acquisition: a sample's explanation-derived score follows the current fitted model and data partition, not a stable domain truth.

Mean EC was 0.4268 ± 0.0363, and mean CF was 0.4680 ± 0.0465 over 25 models. Mean maximum TreeSHAP additivity error was approximately 6.0×10^{-15}. We treat EC and CF as descriptive geometry and additivity as a numerical check; none of them establishes semantic correctness.

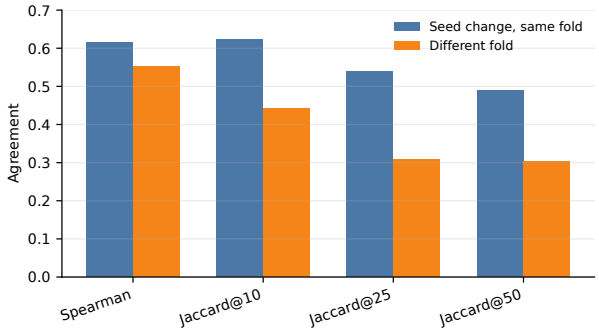


Figure 4: Global and top-k agreement. Fold changes alter the learned attribution ranking more than seed changes within a fixed fold.

7 Discussion

7.1 What the Acquisition Result Means

The clearest result is not that explanation guidance replaces uncertainty. It is that directional attribution balance adds a small, repeatable improvement to

the learning trajectory over random acquisition and a smaller improvement over uncertainty in four of five outer folds. H distinguishes cases in which the same output margin arises from opposing signed masses. Such cases can expose a model boundary supported by competing PE attributes, making them plausible queries for label acquisition.

The algebraic identity in (10) also limits this interpretation. H is determined by the raw-score contribution and the L_1 attribution norm. Likewise, V is determined by L_2 magnitude and the mean signed contribution. A direct margin-plus-norm score may recover part of the measured gain. Until those controls are run, the result supports the tested summaries but does not show that entropy and variance are uniquely necessary.

No policy dominates every objective. $U + H$ leads AULC, $U + V$ reaches the target with the fewest labels, XAI-Full performs between them, and Uncertainty remains strong in restricted-FPR metrics. This is scientifically more informative than a single winner: acquisition policy should be selected against an explicit objective and operating region. A system that values early target attainment may prefer $U+V$, whereas one that values average ranking quality throughout model development may prefer $U+H$.

The target-based label result must also include its accounting boundary. The numbers count labels admitted to model training. Validation labels, the full-pool reference, dataset labeling, and analyst adjudication are additional resources. A 92-label reduction does not imply a measured reduction in analyst hours.

7.2 Batch Diversity and Robust Scaling

The top-100 selection rule can query redundant samples. This risk is material because the batch equals half of the 200-sample initial set. Uncertainty plus k-center selection would test whether the observed gain comes from SHAP signals or whether coverage of the same uncertain region is sufficient. A calibrated committee would test whether ensemble disagreement explains the benefit.

Min–max normalization is also pool dependent. One extreme H or V value can compress the remaining component scores and change the effective mixture. Weight sensitivity, percentile/rank normalization, and robust quantile scaling should be frozen and evaluated before final submission. These studies must use validation or a nested design; selecting the best transformation from test AULC would invalidate the outer comparison.

7.3 What Controlled Deletion Establishes

Top-SHAP replacement consistently damages fitted-model ranking more than uniform random replacement at the same coordinate count, and the result holds for mean and median baselines. This supports a relative model-reliance claim: TreeSHAP ranks coordinates that influence these fitted LightGBM models under the defined intervention.

Several alternatives remain. Tree ensembles use only a subset of the 2,381 coordinates; uniform random masks will often select inactive features. Top-ranked coordinates can also lie farther from their replacement baselines, creating a larger perturbation magnitude and greater distribution shift. A stronger suite would select random controls from model-used features, match absolute standardized displacement, replace semantically coupled feature groups, and compare TreeSHAP with gain/permutation rankings.

An attribution-column permutation is a particularly useful negative control. A common permutation preserves attribution sums, cosine similarities, centroid distances, and norms, so additivity, EC, and CF can all remain unchanged while feature identities are wrong. A valid identity-sensitive pipeline should fail that control. Until it is included, EC and CF are geometry descriptions rather than explanation certificates.

7.4 Stability and Model Maintenance

Moderate within-fold agreement suggests that global attribution summaries are not dominated by model randomness. The larger reduction across folds shows that the learned explanation depends materially on

the sampled data. This matters for model maintenance: a global top-feature list should be versioned with the model and training cohort instead of presented as a timeless malware signature.

The active-learning use of SHAP is local and round specific, so some adaptation of the acquisition ranking is expected. Nevertheless, a deployment study should track rank drift, query composition, and class yield across releases. Sharp changes can signal data drift, feature-extractor changes, or instability that requires investigation before analyst queries are trusted.

7.5 Relationship to Multi-Stage Fusion

The broader thesis studies conditional static/dynamic analysis. A separate 199-sample pilot reported complementary static and API-sequence scores and found that proportional fusion had the highest mean AUC/AP among seven configurations. The PI and PID variants produced identical final decisions; the derivative term added no measurable benefit at 5–20 event prefixes, and the PID-versus-fixed AUC interval crossed zero. Those results motivate evidence-driven simplification, not a full PID superiority claim.

That pilot uses MalBehavD-V1, logistic scorers, and event-indexed API prefixes; the present AL/Fidelity study uses static EMBER vectors and LightGBM. Their metrics cannot be combined into an end-to-end system result. Multi-stage integration requires one paired cohort, a gate with final decisions for all traffic, fixed/learned fusion baselines, and a factorial acquisition–fusion study. We therefore keep fusion outside the contributions and abstract of this paper.

7.6 Practical Implications

The most immediate use is a model-development workflow. Train a static PE model, score a candidate pool, apply a policy chosen for the development objective, present the selected rows to an authorized analyst, and retain query/model lineage. Controlled deletion and stability checks can run as release diagnostics for candidate models. They do not authorize automated remediation or substitute for malware-analysis containment and labeling governance.

The query-computation overhead is modest in the measured small pool, but scaling TreeSHAP to large corpora can become material. A production design may prefilter by uncertainty and calculate explanations only for a candidate band, or cache tree contributions for repeated policy comparisons. Such engineering changes must be re-evaluated because they change the acquisition population.

8 Threats to Validity

8.1 Construct Validity

The label denotes generic benign/malware status, not ransomware. SHA-256 identity and source-index linkage improve traceability but cannot create absent family or behavior semantics. We therefore avoid ransomware, zero-day, and behavioral claims.

H is entropy of signed attribution mass, and V is variance across one attribution vector. Neither is calibrated model uncertainty. The term "explanation-guided" refers to how the signals are computed, not proof that they encode human-understandable malware mechanisms.

Deletion measures model behavior under feature replacement. It does not establish valid PE edits, causality, robustness to every perturbation, or human usefulness. EC, CF, and additivity examine different constructs and are not combined into an OFS.

8.2 Internal Validity

Outer test partitions are excluded from query and target selection according to the retained protocol. Policies share initial sets within fold/seed, limiting confounding by starting labels. The fixed policy weights and min–max normalization, however, were not evaluated through an independent outer sensitivity study. The measured comparison may depend on those design choices.

The oracle reveals existing labels without error. Real analysts can abstain, disagree, or use different evidence, changing both label quality and cost. Batch selection has no diversity constraint and may contain redundant samples.

Mean/median baselines are calculated from training data, avoiding direct test leakage. Yet top-SHAP and random masks are not matched by displacement magnitude, used-feature status, or manifold distance. These confounders can contribute to the deletion gap.

8.3 Statistical Conclusion Validity

Five seed trajectories are averaged within each of five outer folds. The paired outer fold is the comparison unit; fold–seed combinations are not 25 independent test datasets. Five folds support effect-size and direction reporting but limited formal inference. The readiness snapshot does not provide fold-level vectors in this checkout, so figures assembled here omit uncertainty bars. The submission artifact must regenerate them directly from retained fold-level files.

Twenty random masks per k characterize mask variability conditional on each fitted model. The 25 fidelity models share data and folds, so model-level records are also dependent. Pairwise rank comparisons share models and cannot be counted as independent observations in a naive significance test.

Restricted-FPR estimates use small outer test folds. With roughly 300 benign rows, one false positive changes FPR by about 0.33 percentage points. Results at 1% and 5% caps do not validate operation at 10^{-3} or 10^{-4}.

8.4 External Validity

The 2,000 rows form a short-period subset recovered from EMBER material and do not reproduce the full official benchmark protocol. Random row folds do not enforce temporal, family, source, or campaign separation. Results may change under newer software, another organization, a different benign distribution, another feature extractor, or adversarial manipulation. External and temporal validation are required for a broad journal claim.

LightGBM and TreeSHAP are central to the experiment. The acquisition statistics can be defined for other additive attributions, but their numeric behavior, cost, and stability may differ for raw-byte networks, graph models, or sequence models.

8.5 Reproducibility Validity

The readiness report names source commit `fed4322f58f5bb55ec47833e857370d076592a32`, locked dataset/manifest hashes, and replay counts. That commit and the confirmatory result directory are not available in the cloud checkout used for this manuscript revision. We therefore attribute numerical values to the retained readiness report and do not describe this writing pass as an independent rerun. Before submission, the exact source and artifacts must be synchronized, hashes verified, replay executed in a clean environment, and all tables/figures regenerated from source records.

Computational replay establishes consistency with stored data and code. It does not prove external label correctness, data-sharing rights, or generalization.

9 Conclusion

This paper evaluated explanation-guided active learning and controlled TreeSHAP diagnostics for static Windows PE malware classification. The acquisition family combines predictive uncertainty with directional entropy and signed-attribution variance. On an identity-audited 2,000-sample EMBER subset, evaluated with five outer folds and five seeds per fold, $U + H$ achieved the highest ROC/AP learning-curve area. Its improvements over Random were positive in every outer fold; improvements over Uncertainty were smaller and occurred in four of five folds. $U + V$ reached the predeclared validation target with the fewest mean training labels, while XAI-Full reduced the mean budget by 92 labels relative to Random. No configuration was best across learning quality, target attainment, restricted- FPR behavior, and computation.

The TreeSHAP study extended the evaluation to 25 fitted models. Top-ranked coordinates produced

substantially larger AUC decreases than equal-count random coordinates under both training-mean and training-median replacement. Global and top-k rankings showed moderate agreement and were less stable across data folds than across seeds within one fold. These findings support relative reliance of the fitted models on the ranked coordinates under the specified interventions. They do not establish causal malware features or valid executable edits.

The results motivate a focused next experiment rather than a broader deployment claim. Uncertainty plus diversity, calibrated committee acquisition, weight and normalization sensitivity, stronger deletion controls, and external/temporal evaluation are the primary evidence gates. Completing them would determine whether the measured gains reflect uniquely useful explanation signals and whether they transfer beyond the audited subset.

The study also illustrates a general reporting principle for security ML: label efficiency, operating-point quality, attribution behavior, and reproducibility answer different questions. Reporting their trade-offs and limitations provides a stronger basis for model maintenance than selecting a single favorable AUC or explanation image.

A Active-Learning Replay Checklist

For each outer fold and AL seed, a reproducible run should retain:

1. development, validation, and test row identifiers;
2. the shared 200-row initial labeled set and ordered remaining pool;
3. model configuration, library versions, and all deterministic seeds;
4. validation/test probabilities and labels for every round;
5. pool probabilities, $U/H/V$ components, normalized values, composite scores, ranks, and queried row identifiers;
6. the full-pool validation reference and target calculation;
7. per-round ROC–AUC, AP, low-FPR summaries, and scoped timings;
8. saved checkpoints or a deterministic reconstruction mechanism;
9. verification output confirming query membership, uniqueness, no test/validation leakage, additivity, and metric recomputation.

Every policy within a fold/seed must begin from the same initial set. The test partition must remain unavailable to query selection, target construction, weight selection, threshold selection, and stopping.

B Fidelity Replay Checklist

For every outer fold and model seed:

1. retain the fitted model, split IDs, training mean and median, and raw TreeSHAP matrix including the bias separately;
2. verify raw-score additivity before ranking features;
3. save top-k and random mask indices for every sample, k, control draw, and replacement baseline;
4. reconstruct masked matrices from the original test vectors and saved indices rather than from plotted summaries;
5. retain original/masked probabilities and recompute AUC decreases;
6. retain global rankings and model-pair identifiers used for Spearman and Jaccard summaries;
7. label EC and CF as geometry diagnostics and confirm that no composite OFS is introduced without a preregistered rule.

Future controls should include random selection restricted to model-used features, matched standardized displacement, attribution-column permutation, and schema-aware group replacement.

C Interpretation Boundaries

D Claim–Evidence Map

E Pre-Submission Evidence Gates

The following items should be completed before journal submission:

1. synchronize commit `fed4322f58f5bb55ec47833e857370d076592a32` and the complete confirmatory result directory; verify the locked dataset and manifest hashes;
2. rerun the verification entry point in a clean environment and generate every table/figure directly from retained fold-level data;
3. add uncertainty-plus-diversity and a calibrated committee baseline;
4. add weight/normalization sensitivity and batch-redundancy analysis;
5. add identity-sensitive and magnitude-matched fidelity controls;
6. obtain external, temporal, or source-disjoint evidence if the target venue expects broad generalization;
7. verify bibliography metadata against publisher records and confirm authorship, ORCID, corresponding author, data rights, and the current journal template.

Table 9: What the reported quantities establish and do not establish.

Quantity	Supported interpretation	Unsupported interpretation
ROC/AP-AULC	Ranking quality integrated over the measured label budgets	Analyst labor, calibration, or external generalization
Labels to target	Training labels to a fold-specific validation target	Total annotation cost or hours saved
Low-FPR summaries	Relative performance at 1%/5% caps in the outer folds	Reliable FPR at 10^{-3} or 10^{-4}
H	Balance of positive/negative TreeSHAP mass	Predictive entropy or causal conflict
V	Dispersion across signed attribution coordinates	Ensemble or posterior uncertainty
Top-k deletion	Fitted-model sensitivity under training-statistic replacement	Valid PE transformation or causal malware mechanism
Additivity	Numerical reconstruction of the raw tree score	Correct feature identity or human usefulness
EC/CF	Geometry of retained attribution vectors	Explanation correctness certificate
Seed/fold agreement	Sensitivity to fitted model and data split	Stability across organizations and time
Replay verification	Consistency with retained code/data/artifacts	Independent external replication or data authenticity

Table 10: Central claims and submission status.

Claim	Status	Scope or required wording
$U+H$ exceeds Random in ROC/AP-AULC	Reported supported	Five outer folds; report paired effect and direction
$U + H$ universally exceeds Uncertainty	Not supported	Small effects; favorable in 4/5 folds
XAI-Fullsaves 92 training labels vs. Random	Reported supported	Validation labels/reference cost are additional
XAI-Fullis the best policy	Not supported	$U + H$ leads AULC; $U + V$ uses the fewest labels
Top-SHAP deletion exceeds random deletion	Reported supported	Feature space; 25 models; mean/median; absolute effects
TreeSHAP identifies causal malware features	Not supported	No executable-valid or causal intervention
Rankings are invariant	Not supported	Moderate agreement; lower across folds
The study detects ransomware/zero-day families	Not supported	Generic PE malware labels only
The cloud revision independently replayed results	Pending	Synchronize named source/artifacts and rerun verification

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